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Corpus for Orthographic and Emotional Factors in Word and Nonword Recognition

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Datasets:

Repository Name	Dataset Title	Accession Number or DOI	URL to data record	Private reviewer access URL/code
OSF	EmoTL_database.csv		https://osf.io/zpbjy	

Corpus for orthographic and emotional factors in word and nonword recognition

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Abstract

The present article describes a Spanish lexical decision dataset designed to support analyses of how orthographic letter-position coding relates to affective properties during visual word recognition. The dataset provides trial-level response times and accuracy for 406 five-letter words and three transposed-letter nonwords derived from each base word. It also includes lexical, orthographic, and affective measures, including word frequency, orthographic neighborhood, valence, arousal, and discrete emotion ratings. Data were collected online from 92 native Spanish speakers. We describe the materials, procedures, preprocessing steps, and quality checks, and provide technical validation based on well-documented word-frequency and orthographic-neighborhood effects. The repository includes raw data, processed data files, analysis code, and a complete variable dictionary, offering a reusable resource for research and modeling how letter-position coding and affective factors interact during visual word recognition.

Background & Summary

In the last few decades, large-scale datasets in which the number of items or participants is substantially larger than in traditional laboratory-based experiments on word reading have driven both empirical and theoretical progress. The first generation of such studies focused on well-established experimental paradigms that examine core variables known to influence word recognition, as demonstrated by standard laboratory-based research. The English Lexicon Project¹ pioneered this approach, using the prevailing language in cognitive sciences (English) in classic tasks (i.e., lexical decision and naming) to investigate linguistic factors such as word frequency and length, variables directly relevant to the cognitive processes being measured. A second generation of mega-studies expanded to include languages other than English (e.g., Dutch²³; French⁴; British English⁵; Spanish⁶; Portuguese⁷). These studies have not only focused on how different variables affect the recognition of words, but also, although to a lesser extent, on the processing of nonwords⁸. Importantly, nonwords are particularly useful for examining the role of sub-lexical and lexical features, and for identifying the cognitive mechanisms involved in word recognition, which are instrumental in developing theories and models of word processing⁹.

The present work builds on this literature by providing a dataset derived from a large-scale lexical decision study with tight experimental control, in which each base word was used to generate three types of transposed-letter nonwords, and aims to bridge two areas of cognitive psychology that have traditionally been explored in separate research streams, each with distinct theoretical focuses: orthographic processing, such as letter position coding, and

semantic processing, such as affective and emotional properties. To set the stage, a brief description of the current state of these two traditions follows.

A primer on letter position coding

In orthographic processing, the key finding driving both empirical and theoretical work over the last two decades is the transposed-letter effect, which refers to the very robust finding of nonwords formed by transposing two letters, like JUGDE, which comes from the word *JUDGE*, being more easily confused with their base words than control nonwords formed by replacing these two letters (e.g., JUPTE). The transposed-letter effect persists, although slightly weaker, for nonadjacent transpositions (e.g., CHOLOCATE^{10–12}), and it is also weaker when transpositions involve the initial letter¹³. Furthermore, transposed-letter effects are not merely an experimental curiosity due to laboratory word identification tasks, but they are also robust during sentence reading¹⁴.

The transposed-letter effect suggests that individual letter positions are not coded precisely during early reading stages. Several accounts have been proposed to explain this effect, including open-bigram models^{15, 16}, and perceptual accounts^{17–19}. Strong versions of either position seem hard to defend, as evidence supports contributions from both perceptual and representational mechanisms^{20–24}. Hence, researchers from both perspectives agree that the encoding of order within words seems to involve different loci^{21, 25}, and hybrid models do exist²⁶. Thus, further research on the factors modulating this effect is necessary to elucidate the cognitive mechanisms behind orthographic processing.

A primer on the impact of affective properties on word processing

A growing body of research has focused on how affective features influence word processing^{27–30}. Most studies adopt a dimensional approach to emotions, as proposed by Russell³¹, which conceptualizes an affective space defined by two independent dimensions: *valence* (the hedonic tone of a word, ranging from negative/unpleasant to positive/pleasant) and *arousal* (the perceived intensity of a word, ranging from calming to exciting). The effects of emotion on word processing have been reported at multiple levels of processing, ranging from phonology to semantics^{32–37}. Notably, emotion appears to modulate orthographic processing. Longer response times have been observed in lexical decision masked priming tasks for neutral target words with a negative-valence orthographic neighbor than for those with a neutral orthographic neighbor^{38, 39, 40}. These findings suggest the existence of a feedback mechanism from semantics to orthography⁴¹, highlighting the potential for interactions between emotional and orthographic factors.

Notably, previous research has shown that nonwords can convey emotional meaning^{42, 9}. More importantly for the present work, studies have demonstrated the activation of emotional features during nonword processing. For instance, lexical decision responses to nonwords derived from emotionally intense words (e.g. IRTUS from *ICTUS*) are categorized more slowly as 'nonwords' than those derived from neutral words (e.g. CAMELTO from *CAMELLO* [camel]⁴³).

Overview of the study

Nonwords have provided valuable insights into orthographic, lexical, and affective processes. However, the nature of the interaction between emotion and orthographic processing remains poorly understood. Progress in this area depends on research resources

that enable precise control of the materials; to this end, we present data from a lexical decision study that collected response times and accuracy for 406 five-letter words and 1,218 nonwords, representing a larger set of stimuli than is typical in standard laboratory-based studies. These nonwords were generated via internal transposition of letters at three distinct positions: adjacent letters 2-3 and 3-4 (e.g., SLASA and SASLA, respectively, from *SALSA* [sauce]) or nonadjacent letters 2-4 (e.g., SSLAA). This design, in which three nonwords were generated from each base word, enables within-item comparisons across transposition types while holding base-word characteristics constant. Such comparisons are difficult to implement in existing lexical decision datasets, which typically include a single nonword per base word or do not systematically vary the position of the transposed letters^{1, 4, 2, 5, 7}. The restriction to five-letter words and to three predefined transposition types necessarily limits the ecological breadth of the dataset and calls for caution when generalizing the observed effects to longer words or more naturalistic reading scenarios. Nonetheless, the resulting dataset can provide valuable insights into the cognitive mechanisms underlying the processing of orthographic and semantic information in visual word recognition. Furthermore, results from the examination of this dataset can serve as a foundation for extending research to more complex, real-world scenarios in future studies.

This study was conducted online in Spanish, a language spoken by approximately 500 million people worldwide. Using Spanish offers practical advantages; for instance, 1) orthographic processing has been extensively studied and the transposition effects are well-documented¹⁰; 2) there are several comprehensive norming resources that provide emotional and arousal ratings of the stimuli⁴⁴; 3) it has a transparent orthography, like many Western

languages—this contrasts with English, which has often been called an outlier orthography⁴⁵. These features make Spanish an ideal testbed for generalizable insights into visual word recognition.

The present dataset includes a rich array of orthographic, lexical, and emotional variables associated with the items, along with the response times and accuracy data empirically collected for each item and participant. For words (which are also the base-words of the nonwords) we report, among others, frequency per million and movie-based contextual diversity, an orthographic similarity measure (i.e., Levenshtein's distance, presented as the mean distance of the 20 nearest neighbors [i.e., OLD20⁴⁶]), familiarity, imageability, concreteness, age of acquisition, as well as ratings on valence, arousal and several discrete emotions. For each stimulus (word or nonword), we include variables such as the presence of repeated letters, bigram frequency, the number of orthographic neighbors, and consonant–vowel structure (see Table 1 for a summary of all the variables included in the dataset, and the Supplementary Information document for their full description), supporting a wide range of psycholinguistic analyses. In addition, participants' gender, age, and university enrollment level are included to enable researchers to examine potential participant-related differences.

In the following sections, we present the methods of the study followed by descriptive statistics, with an emphasis on analyses by item and visualizing two well-established effects as quality control: word-frequency (for words) and number of orthographic neighbors (for nonwords).

Methods

Participants. Ninety-six native Spanish speakers (aged 18–55 years [mean: 24.09]; 54 female, 41 male, 1 non-binary) participated in the study. Participants were students or alumni from the Nebrija University (Madrid, Spain) and the University of Valencia (Valencia, Spain), all of whom self-reported normal or corrected-to-normal vision and no history of neurological disorders. Four participants were excluded from the final dataset and analysis due to poor performance (overall accuracy below 70%, see *Data acquisition and preprocessing section*). The experiment was approved by the Experimental Research Committee of the Nebrija University (code: UNNE-2022-0017) and adhered to the requirements of the Declaration of Helsinki. All participants provided informed consent, were informed that they could withdraw from the study at any time, and they received €5 compensation for their participation.

Materials

Four hundred and seven five-letter Spanish words were selected as word stimuli from the Spanish ESPAL database⁴⁷ (Zipf frequency: $M = 4.14$ [range: 1.54–6.11]; orthographic neighborhood [OLD20]: $M = 1.44$ [range: 1–2]; e.g., *SALSA* [sauce]). These were 5-letter words in Spanish that did not contain diacritic marks, and whose internal structure allowed for transpositions that did not result in the same stimuli (e.g., transposing letters in positions 2 and 4 of the word *LEVEL* would yield identical stimuli). From these words, three nonword stimuli were created by transposing two letters: transposing letters in positions 2 and 3 (e.g., *SLASA*), transposing letters in positions 3 and 4 (e.g., *SASLA*), and transposing letters in positions 2 and 4 (e.g., *SSLAA*). While the stimulus set initially comprised 407 words (and their corresponding nonwords), one item was excluded during data preprocessing (see *Data acquisition and*

preprocessing section) because only its nonword versions were included in the experimental materials due to a procedural error. Hence, the final dataset consisted of 406 words and their three corresponding nonwords. The complete list of variables is available in the online repository (OSF, see *Data Records* subsection), and a summary is provided in Table 1.

Procedure

The experiment was programmed and run using the Gorilla Experiment Builder⁴⁸. As an online experiment, it required a computer (either desktop or laptop) for participation. After signing the consent form and providing demographic data, participants began the experiment. They were informed that, on each trial, they would see letter strings that could be real Spanish words or invented nonwords. Their task was to press “J” for words or “F” for nonwords as quickly and as accurately as possible—note that keyboard response times provide reliable RT measures in online experiments⁴⁹. Each trial had a 500-ms fixation point (+) preceding the stimulus, which remained on the screen until the participant’s response or a 2,000-ms deadline. There were breaks every 203 trials. The presentation order of trials was fully randomized for each participant. A 12-trial practice block preceded the experimental blocks to familiarize participants with the task. The experimental session took approximately 40 minutes. Each participant completed 1,627 experimental trials (406 words and 1,221 nonwords). During preprocessing, the three nonword trials corresponding to the missing base word were removed.

Data acquisition and preprocessing

Response times (from stimulus onset to key press) and accuracy per trial were recorded per participant. Demographic and raw empirical data were exported as a .csv file from Gorilla Experiment Builder.

We conducted quality checks⁵⁰ to verify the integrity and robustness of the data. Specifically, we ensured that: 1) participants performed the task correctly by removing those with overall accuracy below 70% (i.e., 4 participants, 4.17% of the data); 2) each version of each item had one response per participant—given that the word version of one base item was missing due to a procedural error, the three corresponding nonword trials were removed from the final dataset (0.18% of the data); and 3) all participants completed the same number of trials. The final dataset contains 406 words and 1,218 nonwords, making 149,408 data points (37,352 per condition).

Data Records

The dataset is accessible through the Open Science Framework repository⁵¹. It includes three .csv files: the complete trial-level dataset, EmoTL_database.csv; a word-only subset, EmoTL_Subset_Words.csv; and a nonword-only subset, EmoTL_Subset_NonWords.csv. The two subset files contain observations already included in the complete dataset and are provided to facilitate reuse. Each row corresponds to one trial-level observation; no aggregated data are included. Each column corresponds to a variable, summarized in Table 1 and fully described in SupplementaryInformation_DataDictionary.pdf. We also included a folder (i.e., *Data wrangling and Technical Validation scripts*) that contains the scripts used to generate the dataset (see the

Code availability section), as well as the raw files containing the empirical data and the word properties.

Technical Validation

To ensure the data quality, we first performed descriptive statistical analysis (see Table 2) to examine the distribution of responses and verify that central tendency measures align with previous studies. Note that descriptive statistics exclude responses shorter than 250 ms (0.73% of the total) and, for response times, incorrect responses. The fact that response times to nonwords were faster than response times to words may appear to conflict with the usual lexicality effect. However, changes in word/nonword proportions induce criterion shifts in the responses⁵²: As nonwords were more frequent than words in the present study, the response criterion may have shifted toward a “nonword” response, producing faster nonword responses but lower word accuracy. Moreover, in our design all nonwords were generated by transposing letters from the word stimuli used within the experiment, which may have further contributed to the observed pattern.

Within the nonword data, adjacent letter transpositions incur greater costs than nonadjacent transpositions in both response times and accuracy rates, consistent with prior research^{53, 54}. We also conducted exploratory data analysis (EDA) in R⁵⁵, to confirm that the dataset exhibits two well-established effects on response times and accuracy: for word data, the effect of word-frequency⁵⁶; for nonword data, the effect of number of orthographic neighbors⁸ (see Figs. 1 & 2).

To complement the visual EDA, we fitted linear mixed-effects models in R⁵⁵ (version 4.5.1) using the lme4⁵⁷ package (version 1.1.37) together with the lmerTest⁵⁸ package (version

3.1.3) to obtain the p -values. For word trials, response times for correct responses, inverse transformed ($-1000/RT$) to reduce the positive skew, were modeled as a function of Zipf frequency using the *lmer* function, and accuracy was modeled with generalized linear mixed-effects models using the same predictor and the *glmer* function with a binomial family. For nonword trials, parallel models were fitted using the number of orthographic neighbors as the fixed predictor. Participant and Item number were included as random factors, with random slopes and intercepts when models converged successfully. Full model specifications and scripts are available in the OSF repository. The models showed the expected word-frequency effects for words (RTs: $\beta = -0.09$, $SE = 0.005$, $t = -17.71$, $p < 0.001$; accuracy: $\beta = 0.886$, $SE = 0.06$, $z = 14.816$, $p < 0.001$) and orthographic-neighborhood effects for nonwords (RTs: $\beta = 0.066$, $SE = 0.001$, $t = 88.68$, $p < 0.001$); accuracy: $\beta = -0.395$, $SE = 0.0146$, $z = -27.04$, $p < 0.001$), supporting the technical validity of the dataset.

Usage Notes

The dataset is primarily suited for analyses of item-level and trial-level variation in lexical decision performance. In particular, it allows researchers to compare different types of transposed-letter nonwords derived from the same base words, and to test whether orthographic effects vary as a function of lexical, sublexical, or affective properties. For example, researchers can examine how structural features, such as consonant–vowel structure, affect the processing of transposed-letter nonwords and whether these effects differ based on the type of transposition (adjacent vs. nonadjacent). The dataset also allows for systematic testing of how emotional properties (e.g., valence, arousal, or specific discrete emotions) shape word recognition. Critically, the dataset allows researchers to investigate orthographic–

emotional interactions, an under-explored area that is critical for advancing our understanding of recognition dynamics and for refining computational models of word recognition⁴¹.

Understanding how these characteristics interact to influence lexical access and word recognition is valuable not only for theory-driven research but also for applied domains. For instance, examining the effects of various psycholinguistic and affective factors on word processing can inform the design of assessment tools, intervention strategies, and diagnostic measures for populations with language-related difficulties, such as those with dyslexia or emotional processing deficits⁵⁹. A limitation of the present dataset is that standardized participant-level measures of reading proficiency, vocabulary size, spelling skill, or other individual differences were not collected.

Data Availability

The complete dataset (*EmoTL_database.csv*), together with the word-only and nonword-only subsets (*EmoTL_Subset_Words.csv* and *EmoTL_Subset_NonWords.csv*), is available in the OSF repository (<https://osf.io/zpbjy/files/osfstorage>).

Code Availability

The scripts used to wrangle the data and merge the different datasets to create EmoTL, as well as the code used for the Technical Validation, are available in the OSF repository (<https://osf.io/zpbjy/files/osfstorage>). This work was conducted using R (via the RStudio IDE), version 4.5.1. There are no restrictions on access to the code.

Author contributions

A.B. carried out the data organization, preparation and analysis, and was the primary writer of the manuscript.

J.A.H. contributed to the original idea and contributed to the preparation of the manuscript.

M.P. contributed to the original idea, ran participants and contributed to the preparation of the manuscript.

P.G. contributed to the original idea, carried out data organization, preparation and analyses, and contributed to the preparation of the manuscript.

J.A.D. developed the original idea for the study, created the materials and experimental set-up for data collection, ran participants, and contributed to the preparation of the manuscript.

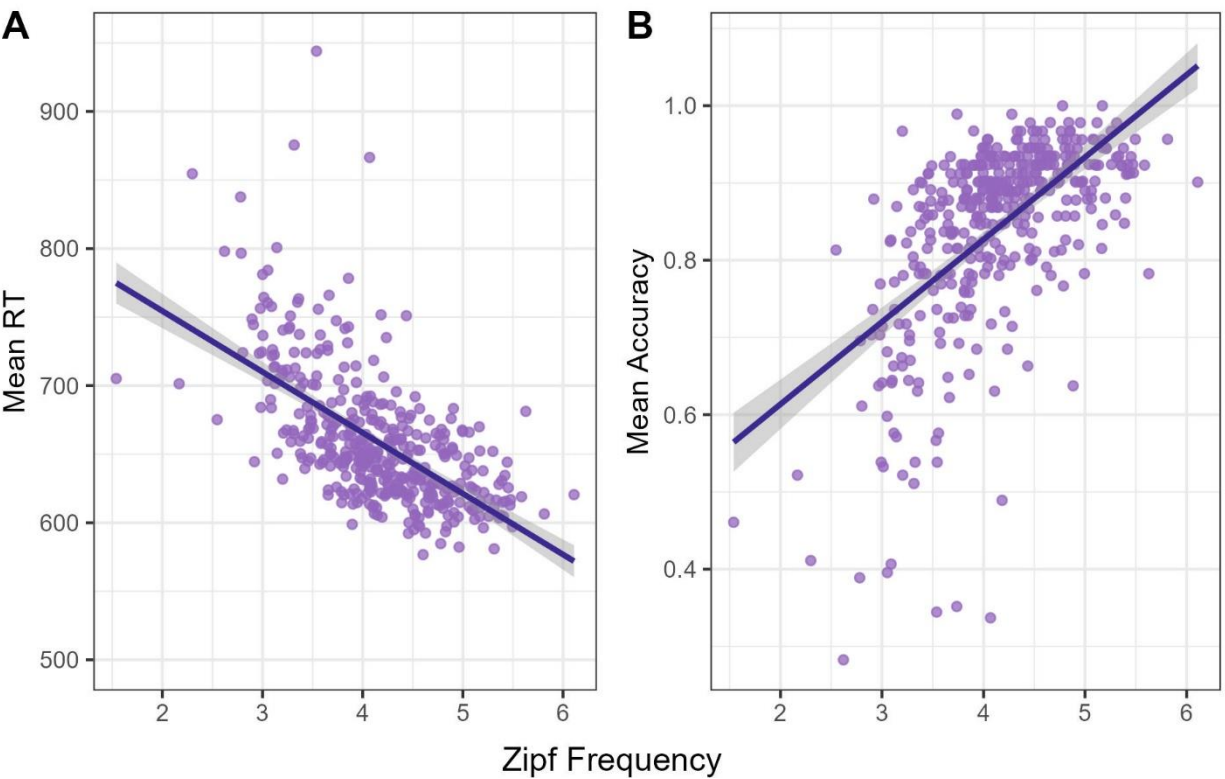
All authors approved the final draft before submission.

Competing interests

The authors declare no competing interests.

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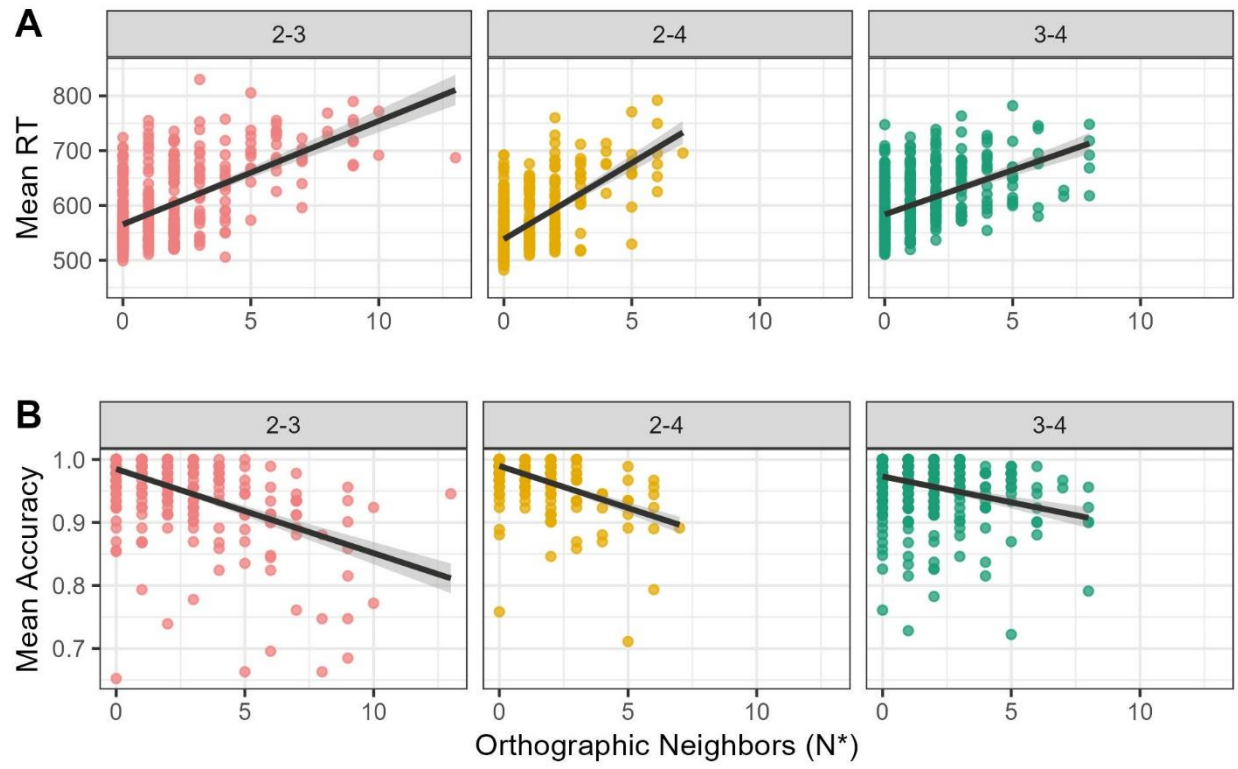
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294 **Figure Legends**

295 Figure 1. EDA visualization of words' mean correct response times (Panel A) and accuracy (Panel
296 B) as a function of the Zipf frequency of the word. Each dot represents a word.

297 Figure 2. EDA visualization of nonwords' mean correct response times (Panel A) and accuracy
298 (Panel B) per type of transposition as a function of the number of orthographic neighbors (N^*)
299 of the item. Each dot is an item.

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301 Tables

302 **Table 1.** Summary of the variables in the dataset—note that for nonwords, the variables that
 303 relate to word properties refer to their base word.

Variable Name	Meaning
Participant.ID	Participant number
Participant.gender	Self-reported gender of participant
Participant.age	Self-reported age of participant
Participant.enrollment.level	Self-reported current study level of participant
TrialN	Trial number
Reaction.Time	Response time
Accuracy	Accuracy
Item	Item (word or nonword presented in the trial)
Condition	Experimental condition (original word, TL 2-3, TL 3-4, TL 2-4)
ItemN	Item number (item identifier shared across all four conditions for a given word)
Rep_letters	Repeated letters in the item (1 =yes, 0 = no)
MLBF	Mean Log Bigram Frequency
CV_O	Consonant-Vowel orthographic structure
Nplus	Orthographic neighbors (number of words differing by one letter [replaced, addition or deletion letter] and by transposing two letters)
Lev_N	Levenshtein N (Levenshtein-based orthographic similarity measure, as implemented in ESPAL, calculated using the OLD20 procedure).
frq	Frequency of use per million
zipf	Zipf scale (frequency measure scaled logarithmically)
CDM_cnt	Contextual diversity (movie-based) count
CDM_frq	Contextual diversity (movie-based) frequency. Percentage.
max_lem_cat	Most frequent part of speech category
syll_structure	Syllabic structure
familiarity	Familiarity (how well-known the word is)
imageability	Imageability (ease with which the word evokes a mental image)
concreteness	Concreteness (how tangible or perceptible the word is)
Valence	Valence (emotional positivity or negativity)
Arousal	Arousal (emotional intensity)
Happiness	Happiness rating
Disgust	Disgust rating
Anger	Anger rating
Fear	Fear rating
Sadness	Sadness rating
AoA	Age of acquisition (the typical age at which the word is learned)
Iconicity	Iconicity (the degree to which the word's sound reflects its meaning)
SERs	Sensory experience ratings
baseword	Base word (for nonwords, the real word from which they were derived)

CVchange	Consonant-vowel change (for nonwords, the specific modifications to the word's consonant-vowel pattern)
CV_comp	Consonant-vowel comparison (between original and TL)

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307 Table 2. Descriptive statistics for the words and the three types of transposed-letter nonwords
308 excluding response times faster than 250 ms. Standard deviation in parentheses.

	Mean RT correct	Median RT correct	Mean Accuracy (%)
Word	653 (148)	621	84.20 (36.5)
TL 2-3	595 (183)	549	96.20 (19.0)
TL 3-4	603 (183)	558	96.25 (19.0)
TL 2-4	558 (160)	518	97.99 (14.0)

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